

Independent Evaluation Report

Human Natural Structure (HNS)
External Verification Proof-of-Concept (PoC) v1.0

Technical Assessment of Structural Auditability,
Failure Taxonomy, and AI Safety Implications

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Framework	Human Natural Structure (HNS) / External Verification Architecture (EVA)
Related Document	HNS Origin & Theoretical Foundation — Volume 1 (Foundational Series)
Status	Proof-of-Concept Evaluation For Standards Discussion & Independent Review
Date	13 June 2026

This report provides an independent technical evaluation of the HNS External Verification PoC. It assesses methodological rigor, alignment with the theoretical framework in Volume 1, strengths in auditability, and requirements for further validation before broader claims can be substantiated.

Executive Summary

This evaluation examines the **Human Natural Structure (HNS) External Verification Proof-of-Concept (PoC) v1.0**, which implements the Multi-Axis Verification (MAV) architecture described in *HNS Origin & Theoretical Foundation — Volume 1*. The PoC introduces an external structural verification layer (HNS-36/144/864 cellular matrix + SMS-6 grounding protocol) operating alongside a base large language model without modifying its weights.

Primary Finding: The PoC successfully demonstrates that HNS can produce an **externally inspectable and structurally attributable record** of how AI outputs are constrained, attenuated, or blocked. This is a significant architectural contribution to AI safety, addressing the opacity problem inherent in current internal alignment methods (RLHF, prompt guardrails).

Performance Observation: Within the tested configuration (single 70B open-source model, 50-turn sequences), the HNS-constrained condition reduced the incidence of five framework-defined structural failure types to 0.0%, compared to 43.1% (baseline) and 28.9% (conventional guardrails). However, this zero-incidence result must be interpreted with caution due to framework-internal scoring and limited sample scope.

Key Strength: The standout achievement is **auditability**. Each intervention is logged with explicit HNS coordinates or SMS layer references, transforming failure handling from opaque probability adjustment into explicit structural verification. This directly supports requirements in ISO/IEC 42001, the EU AI Act, and emerging conformity assessment frameworks.

Critical Caveat: The evaluation methodology carries a risk of circularity because failure categories are both defined and judged within the HNS framework. Independent blind annotation, inter-rater reliability measurement, and cross-model replication are required before performance claims can be generalized. The PoC report itself appropriately frames its strongest defensible claim around auditability rather than universal hallucination elimination.

Overall Assessment: The PoC represents a promising architectural direction. It moves AI safety discourse from "internal model improvement" toward "external structural verification." With rigorous independent validation, HNS has the potential to serve as a candidate module for third-party audit, certification support, and standards-oriented governance logging.

1. Introduction and Purpose of This Evaluation

The HNS PoC was developed to test whether an external structural verification layer, grounded in the Multi-Axis Verification (MAV) hypothesis from cognitive neuroscience, can detect, classify, and constrain specific classes of structural coherence failures in LLM outputs that are not adequately addressed by scaling, RLHF, or conventional guardrails alone.

This independent evaluation was commissioned to:

- Assess the methodological rigor and reproducibility of the reported PoC results.
- Evaluate alignment between the PoC implementation and the theoretical claims in HNS Volume 1 (Chapters I–VI).
- Identify strengths that can support standardization proposals (CEN-CENELEC, JISC, ISO/IEC JTC 1/SC 42).
- Highlight limitations and required next steps for credible external validation.
- Provide constructive recommendations for strengthening the evidence base ahead of broader dissemination or standards engagement.

The evaluation is based on the provided PoC report document and cross-referenced against the extracted content from *HNS Origin & Theoretical Foundation — Volume 1* (particularly Chapters I, II, V, and VI).

2. Overview of the HNS External Verification PoC

2.1 Architectural Foundation

The PoC implements the three-axis MAV model from Chapter I of Volume 1:

Axis	Implementation	Function	Primary Failure Types Addressed
Axis 1 (Generative)	Base LLM (next-token prediction)	Statistical generation of high-prior candidates	— (source of generativity)
Axis 2 (Structural/Causal)	HNS Cellular Matrix (HNS-36 / HNS-144 / HNS-864)	Enforces layer continuity, category boundaries, causal admissibility	Layer Jump, Category Ambiguity, Unsupported Causality
Axis 3 (Grounded/Executive)	SMS-6 Protocol (6-layer dynamic grounding)	Real-time grounding across somatic to universal scales	Scope Drift, Metaphor Contamination

Table 1: Three-axis architecture implemented in the HNS PoC (aligned with MAV hypothesis, Vol. 1 Ch. I)

2.2 Failure Taxonomy

The PoC evaluates five structural failure types explicitly defined in the HNS framework and mapped to clinical correlates in neuropsychiatric phenomenology (Vol. 1, Section 1.5). This taxonomy distinguishes structural coherence failures from purely factual hallucination, enabling targeted verification.

Failure Type	Axis	Definition	HNS Suppression Mechanism
Layer Jump	2	Traversal of meaning layers without bridging proposition	HNS-36 + SMS-6
Category Ambiguity	2	Conflation of cognitive/emotional/intentional categories	HNS-144 + SMS-5
Scope Drift	3	Departure from intended scope without signalling	SMS-2 + SMS-3
Metaphor Contamination	3	Figurative expression displaces structural coordinates	SMS-1 + SMS-6
Unsupported Causality	2+3	Causal claim without mechanistic grounding or admissible path	HNS-864 + SMS-4

Table 2: HNS-defined structural failure taxonomy (cross-referenced to Vol. 1 Appendix A worked examples)

2.3 Experimental Design

Three conditions were compared over multi-turn dialogue sequences (50-turn sequences, 150 turns total) using a 70B open-source base LLM without fine-tuning. Annotations were performed by two reviewers. The design focuses on structural coherence across dialogue trajectory rather than isolated single-prompt factual accuracy.

Condition	Description
Baseline	Base LLM generates responses without HNS structural verification
Guardrail	Base LLM operates with conventional guardrail-style constraints
HNS	Base LLM evaluated under HNS structural constraints (HNS-36/144/864 + SMS-6) via pre-decode interception

3. Reported Preliminary Results

The PoC reports the following comparative incidence rates for the five defined structural failure types:

Failure Type	Baseline	Guardrail	HNS Condition
Layer Jump	18.2%	12.4%	0.0%
Category Ambiguity	14.7%	11.8%	0.0%
Scope Drift	21.5%	15.3%	0.0%
Metaphor Contamination	9.3%	8.6%	0.0%
Unsupported Causality	16.8%	13.1%	0.0%
Overall (any type)	43.1%	28.9%	0.0%

Table 3: Comparative incidence of structural failure types (source: HNS PoC Report v1.0, Section 6)

Important Interpretation Note (from PoC report itself): These figures are framework-internal. The zero-incidence result on HNS-defined categories demonstrates that the verification layer can systematically attenuate or block outputs matching its own failure definitions. It does not yet constitute model-independent proof of hallucination elimination. The most secure claim supported by the data is the production of an auditable structural record.

4. Strengths and Technical Contributions

4.1 Auditability as Core Value Proposition

The most significant and defensible contribution of this PoC is the transformation of failure handling into an explicit, machine-readable, and human-inspectable process. Every correction is attributable to a specific cell in the HNS matrix or an SMS layer. This directly addresses one of the central regulatory challenges identified in Volume 1 (Ch. III, Ch. V): the opacity of neural systems where alignment corrections are encoded across billions of parameters with no legible account.

This positions HNS as a candidate architecture for **third-party audit, certification support, and governance logging** — use cases where self-certification by model providers is recognized as insufficient (EU AI Act, ISO/IEC 42001).

4.2 Coherent Failure Taxonomy

The five failure types are not arbitrary surface labels. They map cleanly to the MAV axes and to clinical phenomenology (formal thought disorder, semantic dementia, frontal-lobe confabulation, etc.). This provides a principled basis for future benchmark design, red-team protocols, and evaluator training — a contribution that extends beyond the immediate PoC results.

4.3 Methodological Self-Awareness

The PoC report demonstrates commendable caution. It explicitly distinguishes what is demonstrated (auditability within the tested configuration) from what remains open (generalizability, universal hallucination elimination). This aligns with the epistemic stance articulated in Volume 1's "A Note on Method and Evidence" and Chapter VI. Such restraint strengthens rather than weakens the work for standards-oriented audiences.

4.4 Model-Agnostic Design Intent

Because HNS operates as an external layer (pre-decode interception or post-hoc audit mode), it is not inherently tied to one model family. This is a prerequisite for vendor-independent assurance and procurement scenarios — a point emphasized in the PoC's conformance-oriented interpretation (Level 0–5 staged model).

5. Limitations and Risks Requiring Attention

5.1 Circularity in Evaluation Methodology (Highest Priority)

Failure types are both defined by HNS and used to judge HNS performance. While this is useful for internal validation and framework development, it introduces a risk of confirmation bias. External reviewers cannot independently verify whether an output truly exhibited "Layer Jump" or "Category Ambiguity" without adopting the HNS ontology. **Independent blind annotation by reviewers unfamiliar with HNS categories, combined with inter-rater reliability metrics (Cohen's kappa or Krippendorff's alpha), is essential** before performance claims can be treated as robust.

5.2 Limited Generalizability

The current results derive from a single 70B open-source base model and a modest number of multi-turn sequences. No statistical significance testing, confidence intervals, or power analysis is reported. Cross-model replication (including commercial API models, smaller models, and multilingual settings) and cross-domain testing (law, medicine, finance, scientific reasoning) are required to establish whether the observed pattern holds beyond the tested configuration.

5.3 Interpretation of Zero-Incidence

A 0.0% incidence rate is striking but requires additional context. It is possible that the attenuation/blocking rules are sufficiently strict that they suppress not only the defined failure modes but also some proportion of valid, creative, or contextually appropriate outputs. Future evaluations should include measures of output utility, task performance, and human preference alongside structural failure rates to ensure that reliability gains do not come at an unacceptable cost to capability.

5.4 Operational and Deployment Considerations

The PoC focuses on functional correctness within a controlled setting. Real-world deployment requires measurement of inference latency overhead, memory footprint, and behavior under high-throughput or edge conditions. The hardware root-of-trust aspiration articulated in Volume 1 Chapter V remains unaddressed at the PoC stage.

6. Alignment with HNS Volume 1 Theoretical Framework

The PoC is tightly coupled to the theoretical programme laid out in Volume 1. Key alignments include:

- **Chapter I (MAV Hypothesis):** The three-axis geometry and orthogonality conditions are directly instantiated in the runtime interception logic.
- **Chapter II (HNS Architecture):** HNS-36/144/864 and SMS-6 are implemented as described; the empirical table in Section 2.6 of the book is taken from this PoC.
- **Chapter V (EVA):** Pre-decode mode is realized; the pseudocode in the book matches the interception logic described in the PoC.
- **Chapter VI (Empirical Status):** The PoC report's own caveats section mirrors the book's distinction between demonstrated results, reported-but-not-validated claims, and open theoretical problems.
- **Appendix A:** Worked examples of the five failure types provide concrete illustrations that the PoC operationalizes.

The PoC therefore functions as the first empirical anchor for the engineering claims in Volume 1. Its cautious framing is consistent with the book's methodological stance that "a framework candid about its evidentiary boundaries is harder to dismiss than one that is not."

7. Recommendations for Next Validation Steps

To elevate this PoC from promising internal demonstration to externally credible evidence suitable for standards engagement, the following prioritized actions are recommended (building on the PoC report's own "Recommended Next Validation Steps"):

Priority 1 — Methodological Rigor (Immediate)

- Conduct blind annotation study with independent reviewers who are not HNS experts; measure inter-rater agreement using established reliability coefficients.
- Publish the annotation rubric and decision criteria in machine-readable form to enable external scrutiny.
- Report statistical measures (confidence intervals, effect sizes) alongside raw percentages.

Priority 2 — Generalizability (Short-term)

- Replicate across at least three additional model families/sizes, including at least one commercial API model under controlled conditions.
- Expand domain coverage to include high-stakes areas (medical reasoning, legal analysis, financial advice, scientific hypothesis generation).
- Introduce adversarial prompt suites specifically designed to trigger each of the five failure types.

Priority 3 — Operational & Governance Readiness (Medium-term)

- Measure and report inference latency overhead, throughput impact, and resource requirements under realistic serving conditions.
- Develop and test machine-readable audit certificate formats; validate that external auditors can reconstruct correction rationales from logs alone.
- Map HNS audit outputs explicitly to requirements in ISO/IEC 42001 (risk management, transparency, human oversight), EU AI Act high-risk obligations, and NIST AI RMF.
- Pilot Level 3–4 conformance (runtime attenuation / pre-decode enforcement) in a controlled enterprise or public-sector deployment.

Priority 4 — Standards & Dissemination

- Prepare a concise technical briefing (4–6 pages) derived from this evaluation for submission alongside existing HNS materials to CEN-CENELEC, JISC, and ISO/IEC JTC 1/SC 42.
- Release the PoC code, logging schemas, and annotation materials under an open license on GitHub (03_NSW repository) to facilitate independent replication.
- Consider developing a short "HNS Conformance Levels" white paper that formalizes the Level 0–5 staged model for use in procurement and certification contexts.

8. Conclusion

The HNS External Verification PoC v1.0 constitutes a meaningful architectural advance in AI safety. By externalizing structural verification and making the correction process explicitly attributable to human-interpretable coordinates, it addresses a genuine gap in current alignment practice: the difficulty of explaining, auditing, and governing how model outputs are constrained.

The reported performance (zero observed incidence of the five defined structural failure types within the tested sample) is encouraging but provisional. The primary and most robust claim supported by the current evidence is **not** that HNS eliminates hallucination, but that **HNS can render the process of constraining outputs structurally legible and auditable**. This is a valuable property in its own right and one that current internal alignment techniques struggle to provide.

With the methodological enhancements outlined above — particularly independent blind validation, cross-model replication, and explicit mapping to governance requirements — the HNS framework has the potential to contribute substantively to the emerging ecosystem of external AI verification, runtime audit, and standards-oriented conformity assessment.

The PoC is therefore best understood as a **successful proof of architectural concept** whose next phase should focus on accumulating independent, reproducible evidence sufficient to support its consideration in international standards and regulatory contexts.

Disclaimer: This evaluation is provided as technical input for discussion purposes. It does not constitute formal certification, endorsement, or regulatory advice. All performance claims remain subject to independent replication and validation.